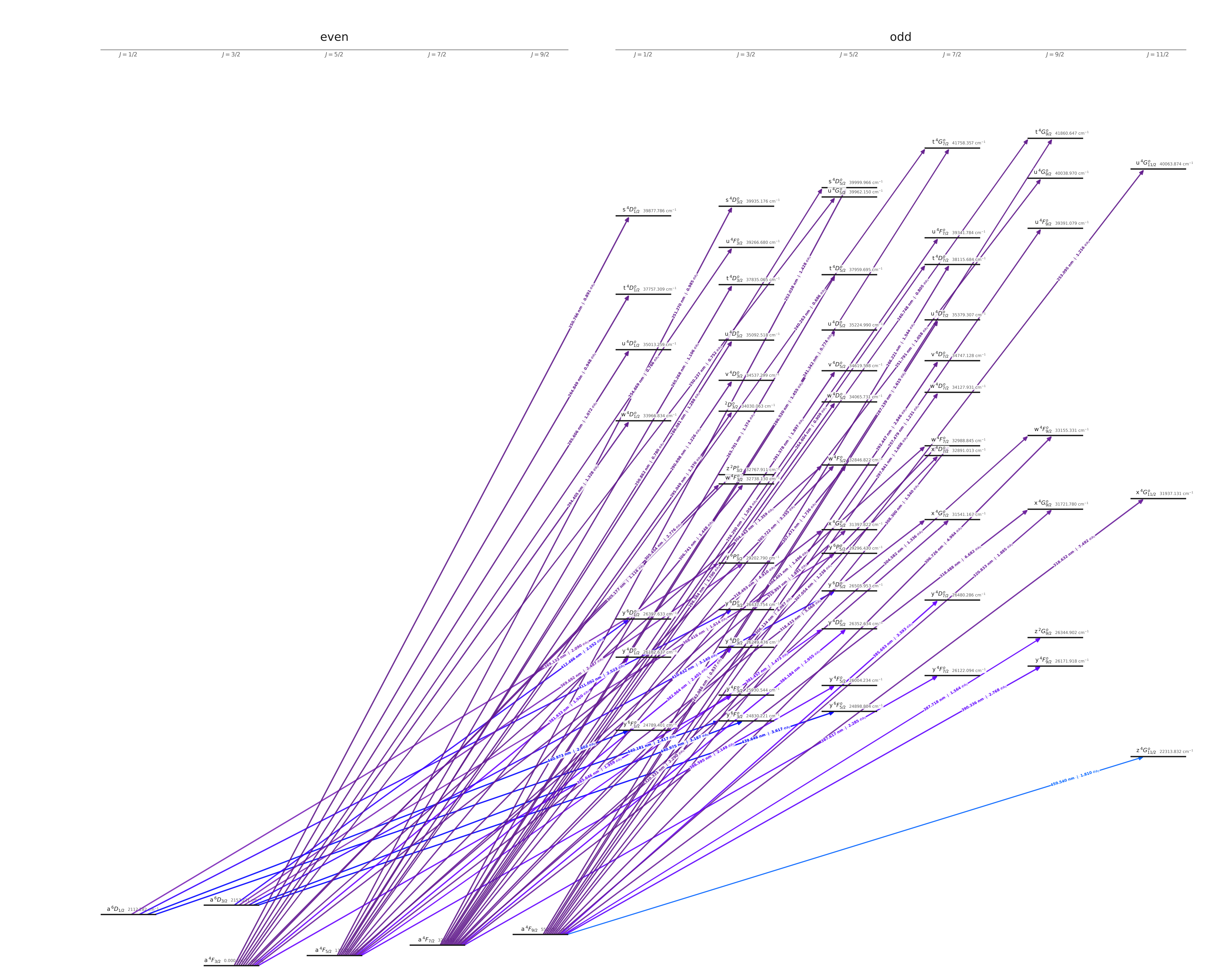


Vanadium

V (V I). Strongest classified transitions below 2 μm (NIST): 60 levels, 70 lines.

V⁺ ionisation limit 54411.67 cm⁻¹ = 6.74619 eV (183.784 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁴ F _{9/2} - v ⁴ D _{7/2} ^o	292.4475	292.3619	1025.1155	2.846	8.200e+07	-	-	
a ⁴ F _{3/2} - w ⁴ D _{1/2} ^o	294.4048	294.3188	1018.3001	1.328	7.000e+07	-	-	
a ⁴ F _{3/2} - w ⁴ F _{3/2} ^o	305.4542	305.3654	981.4645	2.776	1.370e+08	-	-	
a ⁴ F _{3/2} - w ⁴ F _{3/2} ^o	305.7221	305.6333	980.6043	3.355	1.330e+08	-	-	
a ⁴ F _{7/2} - w ⁴ F _{7/2} ^o	306.1342	306.0452	979.2844	4.067	1.460e+08	-	-	
a ⁴ F _{9/2} - w ⁴ F _{9/2} ^o	306.7261	306.6370	977.3946	4.964	1.730e+08	-	-	
a ⁴ F _{3/2} - x ⁴ G _{1/2} ^o	318.4330	318.3409	941.4618	5.668	2.520e+08	-	-	
a ⁴ F _{7/2} - x ⁴ G _{7/2} ^o	318.4881	318.3960	941.2988	6.682	2.800e+08	-	-	
a ⁴ F _{3/2} - x ⁴ G _{3/2} ^o	318.4934	318.4013	941.2830	4.910	2.520e+08	-	-	
a ⁴ F _{9/2} - x ⁴ G _{9/2} ^o	318.6319	318.5398	940.8739	7.492	2.930e+08	-	-	
a ⁶ D _{3/2} - y ⁶ P _{3/2} ^o	369.6917	369.5865	810.9257	2.487	6.200e+07	-	-	
a ⁴ F _{3/2} - y ⁴ D _{1/2} ^o	381.9325	381.8241	784.9357	1.920	6.700e+07	-	-	
a ⁴ F _{9/2} - y ⁴ D _{7/2} ^o	385.6934	385.5840	777.2818	3.593	5.700e+07	-	-	
a ⁶ D _{3/2} - y ⁶ D _{1/2} ^o	412.4662	412.3498	726.8292	2.552	9.400e+07	-	-	
a ⁶ D _{1/2} - y ⁶ F _{7/2} ^o	440.9731	440.8493	679.8429	2.602	8.000e+07	-	-	
a ⁶ D _{3/2} - y ⁶ F _{3/2} ^o	440.9754	440.8516	679.8394	3.187	6.000e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||e||J'⟩| in ea₀.
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).

